

Experiment 6: Basic Video Processing Operations

Aim

To read a captured video using Python and display the video in normal, slow motion, and fast motion.

Algorithm

1. Import the required libraries (cv2, sys, and pathlib).
2. Read the video using cv2.VideoCapture().
3. Set the playback delay for slow, normal, and fast modes.
4. Read and display each video frame.
5. Press 'q' to stop the video playback.
6. Release the video object and close all windows.

Python Code

```
import sys
from pathlib import Path
import cv2

def play_video(video_path, mode="normal"):
    capture = cv2.VideoCapture(video_path)

    delays = {"slow":100, "normal":30, "fast":5}
    delay = delays.get(mode, 30)

    while True:
        success, frame = capture.read()
        if not success:
            break

        cv2.imshow("Video", frame)

        if cv2.waitKey(delay) & 0xFF == ord('q'):
            break

    capture.release()
    cv2.destroyAllWindows()

play_video("input.mp4", "normal")
# Use "slow" or "fast" to change playback speed.
```

Result

The captured video was successfully read using Python and displayed in normal, slow motion, and fast motion.